

Tharun Kumar Bandaru

Data Scientist @ C3.Ai

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Summary

Previous	Data Scientist @ Elephas
Preferred	Data Scientist
Location	San Mateo, CA, US
Desired Work Settings	Remote or On-Site or Hybrid
Willing to Relocate	No
Work Authorization(s):	Now or in the future will require visa sponsorship for employment in the United States.
Employment Type	Full-time
Total Experience	6 years
Education	Bachelors @ JNTU Masters @ GEORGIA STATE UNIVERSITY
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Tharun Kumar Bandaru

| Redwood City, CA |

| [GitHub](#) | [LinkedIn](#) |

SUMMARY

- **Data Scientist with 4+** years of experience in **Healthcare** and **Finance** industry, skilled in developing LSTM, Transfer learning, imaging methods for healthcare research and GLM, Anomaly Detection, Demand Forecasting for finance.
- Data Science and Analytics graduate with a focus on Big Data and Machine Learning with a GPA of **3.9**.
- Experienced in Python, Machine Learning, Data Science, SQL, ETL pipelines, CI/CD, EDA, Azure, AWS. 3x Winner of hackathons, certified in Tableau, Azure AI. Plays with Large Language Models (LLM), QLoRA, LoRA, Vector Databases, Fine-tuning **LLMs**, RAGs, Gen AI.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SAS, SQL, C, Java, HTML, CSS, Javascript, Matlab.
- **Frameworks and ML Skills:** PyTorch, Tensorflow, Keras, Scikit, NLTK, SpaCy, Numpy, JAX, Pandas, MLOps, OpenCV, Time Series Modeling, NLP, Transformers, BERT, Cosmos DB, ETL, LLM, Langchain, LLaMA index, QLoRA, LoRA, RAG, Vector Database. Statistics (Hypothesis testing, ANOVA), Regression (Linear, Logistic, Ridge, Lasso), Decision Tree, Random Forest, XG Boost, K-Means, KNN, Support Vector Machine, A/B Testing, NLP, Scikit, Seaborn, Plotly.
- **Tools and others:** Tableau, Power BI, Spark, Hadoop, Git, GitHub, Docker, RASA, Graph Database, Databricks, PySpark/Spark, Azure, AWS, Streamlit. Jenkins, Docker, CI/CD, Git, Kubeflow, Airflow.

EXPERIENCE

C3.AI, Redwood city, CA

Nov 2024 – Present

Senior Data Scientist

- Migrated data from V7 version of entire project to V8. Handled feature set creation, setting up data workflow. Responsible for data cleaning through outlier detection, connecting with end clients for outlier review.
- Feature engineered and processed 500k rows using Property Data, past appraisal history, location size, age, GIS data and relevant features.
- Built AI-AVM (Automated Valuation Models) in predicting the real estate property appraisal values utilizing decision trees and boosting techniques from **LightGBM** (Light Gradient Boosting) model
- Performed hyperparameter optimization using n-estimators, subsamples, learning rate.
- Evaluated model on custom metrics like Price-Related Differential, Percentage of sales price and using MAPE, and R-square methods.

ELEPHAS, Madison, WI (Remote)

May 2024 – Sep 2024

Data Scientist

- Contributed to medical immunology research at a dynamic startup specializing in advanced imaging and data-driven immunotherapy.
- Utilized MPM (Multi Photon Microscopy) data to evaluate the performance of medical immunology treatment, PCA for dimensionality reduction and Binning to create custom buckets of similar values which improved data analysis efficiency.
- Performed Data analysis on cytokine data to assess the response levels on protein level.
- Experimented with feature selection techniques such as **Recursive Feature Elimination with Random Forest and Gradient Boosting, and Genetic Algorithms with auto encoders and GMM**, leading to improved model performance.
- Applied k-means, DB-SCAN, Hierarchical clustering, and Spectral clustering on assay samples to analyze drug responses based on protein levels, resulting in more accurate clustering of response patterns.

HONEYWELL, Atlanta, GA

May 2023 – Aug 2023

Data Science Intern (Skills: MS SQL Server, Tableau, Python, Hugging Face, Azure Event Hubs, Power BI, Azure Data Factory, Azure Data Lake Storage)

- Analyzed diverse data tables with MS SQL Server within industrial repair solutions, pinpointing faults, errors, and potential improvement opportunities. Leveraged **Tableau** to visualize and communicate insights to cross-functional stakeholders, enhancing collaboration and decision-making processes.
- Employed SAS for advanced statistical analysis and predictive modeling to identify trends and anomalies in repair data, leading to improved maintenance strategies and operational efficiency. Used **SAS Viya** for cloud-based analytics and machine learning, enhancing the scalability and performance of data analysis workflows.
- Utilized Support Vector Machines (**SVMs**) to **detect anomalies in smoke sensor data**, improving the accuracy of fault detection and reducing false alarms. Implemented LSTM models for accurate demand forecasting in critical sensor markets, facilitating proactive inventory management and cost optimization.
- Leveraged Falcon 7B LLM through Hugging Face to build a responsive answering system, delivering data-driven repair suggestions based on historical records, resulting in streamlined customer support interactions.

TRENDS Lab (GEORGIA TECH UNIVERSITY)

Aug 2022 – Apr 2024

Machine Learning Research Assistant - Healthcare (Skills: Python, MATLAB, R language, Statistics, Neuroimaging, LSTM, ICA, U-Net)

- Researched on ICA, brain disorders using fMRIs Functional Connectivity, StaticFNC, DynamicFNC. Designed, analyzed, and visualized high-dimensional time-series data to understand neurobiology for neurological diseases.
- Built a LSTM model for the classification of Alzheimer's (AD) and Schizophrenia (SZ) on rs-fMRI **Time Series**.
- Employed **stratified k-fold** on an imbalanced dataset finding best hyperparameters and model, achieving an accuracy of 80%, sensitivity of 79%, and specificity of 78% with the best model.
- Utilized ensembling methods to improve model performance, resulting in enhanced prediction accuracy for neurological disease classification.
- Presented research findings at **OHBM 2023 in Canada**, contributing to the community's understanding of neurological disease modeling.

- Built a U-Net model leveraging ImageNet models like **ResNet** and **DenseNet** as encoders on mammographic images. Optimized model performance through extensive hyperparameter tuning, achieving an AUC-ROC score of 0.94.

ACCENTURE, Bangalore, India

Jun 2021 – Jul 2022

Data Scientist (Skills: XGBoost, SVR, Regression, Neural Networks, ETL, Time Series, BERT, Azure ML)

- Developed regression models to **predict the time of recovery of a patient and Customer Lifetime Value using Lasso, Ridge, Support Vector Regression (SVR), and XGBoost.**
- Designed and implemented ETL pipelines leveraging **Azure Data Factory** and **Azure Data Lake Storage Gen2** for automated data cleaning and transformation, ensuring high-quality data for model training and analysis.
- Established **CI/CD** pipelines with **Docker** containers for seamless model deployment and monitoring, guaranteeing efficient scaling and reproducibility on **Azure Monitor.**
- Applied machine learning models, including Generalized Linear Models (GLMs) and XGBoost, achieving 85% accuracy in predicting insurance claim costs, enhancing risk assessment precision and financial forecasting reliability.
- Developed an anomaly detection system using K-Nearest Neighbors algorithm, successfully identifying 90% of fraudulent insurance claims, enhancing the company's fraud prevention capabilities.
- Annotated physician notes with corresponding medical codes to train **BERT-based NLP** models for accurate extraction.
- Integrated **BERT-based NLP** models into the claims processing pipeline on Azure **Databricks**, achieving 92% accuracy in extracting medical codes from physician notes, significantly reducing manual workload and errors.
- Built a real-time fraud dashboard using Power BI and Azure SQL Database, providing insights into suspicious activity patterns.

TECHCITI TECHNOLOGIES, Bangalore, India

Nov 2019 – Jun 2021

Data Scientist (Skills: Apache Spark, Apache Airflow, MongoDB, Synapse Analytics, A/B testing)

- Developed a personalized news **recommendation system** using collaborative filtering techniques, including user-based and item-based methods, to offer tailored content suggestions.
- Implemented advanced similarity algorithms (cosine similarity, Pearson correlation) to enhance recommendation accuracy, resulting in a 15% increase in user retention.
- Designed and managed ETL workflows using Apache Airflow, processing large-scale data sets efficiently for model training and updates, Apache Spark for scalable model training and recommendation generation, reducing processing time by 40% for large datasets.
- Implemented a CI/CD pipeline using Azure DevOps for version control and automated model deployment, reducing deployment time by 30% and improving overall development efficiency.
- Integrated MongoDB for real-time updates to the recommendation engine, enabling dynamic content personalization. Integrated Azure Event Hubs to enable real-time data streaming, delivering up-to-date recommendations based on immediate user interactions.
- Conducted rigorous **A/B** testing to optimize recommendation strategies, leading to a 20% increase in article click rates.
- Leveraged **Azure Synapse Analytics** for comprehensive data analysis and Azure Databricks for scalable model deployment and monitoring.
- Incorporated SHAP (Shapley Additive explanations) for interpretability and understanding of the recommendation model's decision-making process, enhancing transparency and trustworthiness in recommendations.

PROJECTS

- **Enhanced LLAMA 2 with Custom Insurance Q&A Dataset:** Successfully fine-tuned the **LLAMA 2** large language model using Low-Rank Adaptation (**LoRA**) and Quantized Low-Rank Adaptation (**QLoRA**) techniques. Leveraged a custom dataset of human-annotated question-answer pairs from an insurance company database, significantly improving model performance on specific domain.
- **Real-Time Face Emotion Detection:** Captured and segmented real-time face emotions video using the **haar-cascade classifier**, implemented deep learning model to classify various emotions such as sad, angry, happy, neutral, and surprised.
- **LLM PDF Chatbot:** Developed an end-to-end chatbot application utilizing the **Mistral-7B** and **Flask**, fine-tuned using TRL and **Hugging Face**. Fine-tuning was performed on the Alpaca dataset to optimize the model's performance for generating contextually responses.
- **Electricity Load(Time Series) Forecasting:** Built ETL pipeline in **PySpark** to ingest data. Developed LSTM model on Databricks.
- **End-to-end Healthcare application:** Integrated **OpenAI-Whisper** for speech transcription, **Gaussian Naive Bayes** for diagnosis, and Haar-Cascade Classifier for heart rate monitoring. The app addressed pain points, such as provider shortage and burnout. **Won 1st place.**
- **Sentiment Analysis on Yelp data:** Performed sentiment analysis using **LSTM** with word embedding and **TF-IDF**. Implemented LDA for comprehensive topic modeling to extract insights from review data.
- **Building a RAG Application Using Python, LangChain, and OpenAI API:** Developed a Retrieval Augmented Generation (**RAG**) system using Python, **LangChain**, and the OpenAI API. Utilized **GPT 3.5 turbo** model for intelligent question-answer generation from video transcripts. Integrated **Pinecone** as a vector storage database for efficient text retrieval. Implemented tokenization, embeddings, and similarity search algorithms for accurate response generation.
- **Fintech AI:** Developed an end-to-end financial analysis tool using **LangChain**, **OpenAI embeddings**, and ChromaDB for embedding and analyzing SEC 10-K filings. Integrated BeautifulSoup for text extraction and Matplotlib for visualizing key financial metrics.

RESEARCH PUBLICATIONS

- Design and analysis of attention-based mechanisms for intent recognition and classification: [[Paper Link](#)]
- A Survey on recent advances in pneumonia detection using ChexNet: [[Paper Link](#)]
- Classification of Alzheimer's and Schizophrenia using rs-fMRI: [[Paper Link](#)].

EDUCATION

Georgia State University - Master of Science in Data Science and Analytics (GPA: **3.9/4.00**)

JNTU Hyderabad - Bachelor of Technology in Electronics and Computer Engineering (GPA: **3.6/4.00**)

ACHIEVEMENTS

- **Winner – Georgia Tech Hackathon 2023- *End-to-end Healthcare application*:** Integrating **OpenAI-Whisper** for speech transcription, GPT-4 for patient interaction, and Haar-Cascade Classifier for heart rate monitoring. Leveraged **BERT for EHR** analysis and Gaussian Naive Bayes for diagnosis, addressing provider shortage and burnout. Achieved 1st place for innovation and impact.
- **Winner – Google Hackathon 2023:** Developed an Appsheet based interactive app to address pain points in healthcare, imaging.
- **Runner-up – Georgia Tech Hackathon 2024-** Developed and fine-tuned a chatbot model using OpenAI's GPT-3.5-turbo and Langchain, processing CSV data into JSONL format while ensuring token and format compliance. Integrated the model with a Gradio UI and **llama** for conversational interactions. Deployed using Docker for both development and production environments.